Available online at www.sciencedirect.com

ScienceDirect

IFAC PapersOnline CONFERENCE PAPER ARCHIVE

IFAC PapersOnLine 55-9 (2022) 12–17

Automatic emergency load control of electric power systems: trends, new solutions, problems

N.I.Voropai*, M.V. Chulyukova, A.A. Petrov*****

**Melentiev Energy Systems Institute, Irkutsk, Russia*
(e-mail: ni.voropai@yandex.ru)

***Far-Eastern Grids Company, Blagoveshchensk, Russia*

(e-mail: margarita_kaigor@inbox.ru)

**** Irkutsk National Research Technical University, Irkutsk, Russia*
(e-mail: artem.petrov11@mail.ru)

Abstract: The article considers the role of automatic emergency load control implemented by various types of emergency automation to ensure the stability and survivability of power systems (PS). A new automatic system is proposed to prevent an emergency shutdown of the load of active consumers during emergency allocation of the power supply system to islanded operation. The problems of coordination of various types of automatic emergency load control systems are formulated, taking into account changes in the principles of their implementation.

Copyright © 2022 The Authors. This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0/)

Keywords: Electric power systems, automatic emergency load control, coordination.

1. INTRODUCTION

Emergency automation (EA) it plays a crucial role in ensuring the stability and survivability of power systems (Sovalov, et al, 1988), (Shulginov, et al, 2009), (Ying-Yi, et al, 2021), etc. The transformation of electric power systems and the appearance of new properties in them, the use of new technologies, including for managing modes, and the increase in requirements of consumers for modes and control actions (CA) on the part make it necessary to improve the EA, give it new functions and capabilities (Koshcheev, 2008), (Voropai, 2020) etc.

One of the most common CA during the operation of the EA is associated with an emergency load shedding, it is used in various types of automation. It is especially necessary to mention the device that historically received the name of special automatic load shedding (SALS), the purpose of which is to limit or remove the overload of the network element (s) and prevent the violation of the stability of the PS by disconnecting a certain amount of load in the receiving part of the system. The SALS is intended to work taking into account the emergency shutdown network or generating element of the PS, which determined its specificity – to work "in advance", i.e. before all other types of EA, as for the latter, as in Russia and other countries of the triggering factor is output over the prescribed limits of the parameters of the electric mode of the system. (Faranda, et al, 2007), (Adamiak, et al, 2006), (LuJinling, et al, 2010). This reduces the performance of such an EA compared to the SALS.

It is impossible not to mention the extremely important role of the CA for automatic emergency under-frequency load shedding (UFLS), as well as in local automatic equipment overload limitation (AEOL) and voltage reduction limitation (AVRL) (Sovalov, et al, 1988).

Taking into account the multiplicity of important functions of emergency control, which are solved with the use of CA for emergency shutdown of the load, many works investigate the possibilities of improving the efficiency of various types of EA that implement the corresponding CA.

Further, in section 2, the most typical research in this area is analyzed, and in section 3 the ideas of a new automatic emergency load control for active consumers (AELCAC) are presented. Section 4 provides case study of AELCAC operation scenario. Section 5 contains some generalizations on the problems under consideration, based on which new problems that need to be solved are formulated. Section 6 provides a conclusion based on the results of the performed studies.

2. MAIN DIRECTIONS TOIMPROVE THE EFFICIENCY OF AUTOMATICEMERGENCY LOAD SHEDDING

Modern requirements for automatic emergency load control are determined by objective factors related to the transformation of internal and external operating conditions of the PS, which determines significant changes in the requirements for the operation of the corresponding automation systems. One of these fundamental requirements is related to increasing the adaptability of automatic emergency load shedding. Let's consider a number of examples of implementing the corresponding algorithms.

In the paper (Danilov,et al, 2014), it is proposed to select in real time (from a pre-calculated set of load values set up for disconnection) the stage in accordance with the impact

2405-8963 Copyright © 2022 The Authors. This is an open access article under the CC BY-NC-ND license.
Peer review under responsibility of International Federation of Automatic Control.
10.1016/j.ifacol.2022.07.003

volumes specified for specific circuit-mode conditions. A different algorithm (Panasetsky,et al, 2011) is based on the use of sensitivity coefficients and changes flow in the active power flow along the line to changes in the active load in the node (or to changes in the flow along another line), while solving the problem of linear minimization of the load turned off by automation in the current circuit-mode conditions.

In the work (Voropai et al, 2016), the principle of adaptive distributed load disconnection by SALS devices refers to disconnecting non-responsible electric receivers of specific consumers in the distribution network, disconnecting concentrated electric-intensive consumers or large energy districts powered from high-voltage substations. At the same time, a new task arises to control overloads in the distribution network, the solution of which leads to a delay in the implementation of CA and the loss of the main advantage of SALS – speed. To eliminate this drawback, the authors (Voropai, et al, 2016) proposed a combined principle of CA implementation under the action of SALS: the high-speed part of the CA is implemented with the lowest possible delay (high-speed stages) which provides rapid unloading of the overloaded direction and there by preventing a violation of the stability of the PS, and then the second part of the CA with a significant possible delay to ensure the critical stability of the PS in the post-accident mode (stages of distributed adaptive SALS). At the same time, the second part of the CA is implemented excessively in order to restore the necessary amount of power to electric receivers that are disconnected by the high-speed part of the CA.

Another important requirement for up grading the principles of automatic emergency load shedding is to transfer the implementation of CA from the level of 6-10 kV to the level of 0.4 kV (Giyoev, et al, 2017). This makes sense from the point of view of more detailed consideration of the energy dependence of the infrastructure, tightened safety requirements, social significance and commercial factors. The main idea is to replace the complete shutdown of 6 – 10 kV feeders with a partial limitation of their load with the possibility of maintaining power to the most important electric receivers. Itis advisable to use this principle not only in relation to the UFLS, as it is proposed in (Giyoev, et al, 2017), but also in the formation of adaptive algorithms for automation of distributed load disconnection, presented in (Voropaiet al, 2016).

However, the implementation of the considered principle faces a number of significant problems. In addition to the loss of performance of CA selection algorithizms due to the need to control overloads in the distribution network, at the voltage level under consideration, large frequency measurement errors are possible in conditions of significant signal distortions, for example. This is especially true for energy districts with a large share of distributed generation, which leads to erroneous decisions on the choice of CA. In order to overcome these problems, (Kulikov,et al, 2021) suggests applying a sequential analysis Wald's procedure. Ensuring guaranteed high performance of the UFLS device is achieved by using adaptive settings generated off-line and allowing you to reduce the uncertainty zone when recognizing the operating mode of the energy district. In addition, from the point of view of ensuring the speed of CA selection, a multi-channel implementation of an UFLS device with a parallel structure of processing and decision-making channels for each UFLS stage is effective.

The review article (Quint, et al, 2019) analyzes the experience of Australia and some US states in real-time load control under different conditions and for different purposes. For example, in Hawaii, for an isolated power supply system, the role of the UFLS is very important, responding to a decrease in frequency during periods of accidental reduction in power output from renewable energy sources. In these situations, the UFLS turns off secondary electrical receivers, thereby reducing the power requirements and energy intensity of electricity storage devices. Another example of real-time load control is the PS in California, USA, where about a third of electricity is generated using renewable energy resources, mainly solar energy. The approach is based on a short-term forecast of power output from solar power plants. If high solar activity is predicted for the period of the daily minimum load, the controlled part of the electric receivers is shifted from the peak part of the load curve to its minimum.

The listed cases of real-time load management in Australia and the United States relate to specific operating conditions of electric power systems. With regard to the widespread use of traditional UFLS in many countries, many authors conduct research on improving the adaptability of this automation, its distributed implementation, etc. (Adamiak, etal, 2006), (Ford, etal, 2009), (Jung, etal, 2002) etc.

In the work (Ying-Yi, etal, 2021), it is noted that there are two categories of automation of emergency shutdown of the load: upon the occurrence of an event (approximately similar to the SALS mentioned above) and in response to a change in the variable of the PS mode (frequency, voltage, transmitted power). The authors study the features of the first category of emergency load shedding automation, which aims to prevent the cascade development of an accident in an isolated PS with a large share of renewable energy sources. Scenarios for peak and off-peak zones of daily load curves in summer and winter conditions with one or two simultaneous failures of system elements were studied. For each scenario, the emergency load shedding was minimized using the particles warm optimization, while parallel computing technology is used to speed up calculations. Studies conducted for a 38-node isolated PS, including wind turbines and photovoltaics, demonstrated the effectiveness of the developed approach.

The analysis given in this section information of the directions of modernization of the automatic emergency load shedding initiated by the new conditions and requirements to their work, allows you to do a number of important generalizations, which will be discussed further in section 4 based on the discussion in the next section, a new automation emergency load control of active consumers (AELCAC), proposed in (Voropai, et al, 2020).

3. AUTOMATION SYSTEM FOR EMERGENCY LOAD CONTROLOF ACTIVE CONSUMERS

The category of active consumers includes those who, when using differentiable electricity tariffs during the day, can independently on-line manage their own electricity consumption by transferring the operation of some electric receivers from the high-tariff zone to the low-tariff zone (Conejo, etal, 2010), (Aizenberg, etal, 2016).

Power systems (PS) with an active consumer differ from traditional systems. Integration into newly formed PS of own energy generation based on unconventional and renewable energy sources, energy-consuming equipment capable of changing or transferring loads over time, energy storage and centralized PS forms new functional properties and technological capabilities for solving not only economic problems, but also emergency management tasks. (Stennikov, et al, 2016).

When allocating a PS with an active consumer to isolated operation in emergency situations, emergency management should be based on the principle of a layered defense system, as is traditionally accepted. At each turn, certain controls are used to stop or weaken the unfavorable development of the emergency process and ensure the transition to a steady state.

As already mentioned, in most cases, automatic shutdown of the consumer load is used to restore the parameters. At the same time, UFLS is the most effective tool. However, the implementation of CA for disconnecting consumers in the 6-10 kV network does not take into account the presence of responsible consumers at the level of 0.4 kV, an interruption in the PS of which is unacceptable. AELCAC automation makes it possible to solve this problem based on the property of electric receivers of active consumers to control their load.

The moment of emergency shutdown of the supply substation and the transfer of the PS to islanded operation is random, and the most severe emergency will occur during the daily maximum load. In this regard, the automatic transfer of some electric receivers by the load management system from the maximum load to the minimum (later) it will allow providing electricity to additional consumers during the daily maximum load period compared to the initial situation and, if necessary, increasing the post-accident frequency level.

The same action can be performed by the proposed automation at another moment of the daily load curve, which is not in the zone of the daily maximum. In any case, the automation system operates upon an emergency shutdown of the substation and the transfer of the PS to islanded operation. Fig. 1 shows the developed block diagram of the device implementing the AELCAC algorithm.

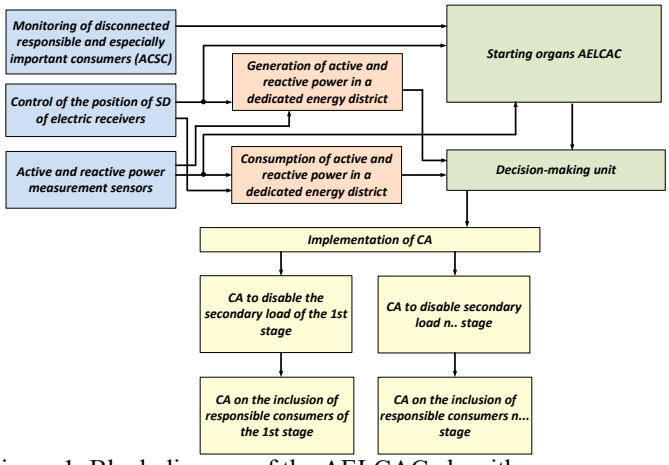


Figure 1. Block diagram of the AELCAC algorithm.

A necessary condition for the functioning of AELCAC is an established balance of generated and consumed power, which can be achieved by turning on the "active consumers" of their own generation. In this case, the frequency and voltage may deviate from the requirements for the parameters in normal mode. Monitoring the current balance of active and reactive capacities in the allocated energy district allows for an optimal choice of CA. Monitoring the positions of switching devices (SD) in a dedicated section of the energy district makes it possible to determine the state of the electrical receivers and the presence of voltage in them.

Based on the information received about the power balance and the disconnected responsible consumers, the AELCAC operation algorithm is formed, which activates the block of "Starting Organs AELCAC". After processing the results of all measurements by the "Decision-making unit", control actions are formed to turn off the secondary load and turn on the load of responsible consumers. Switching on the load of responsible consumers is carried out using existing automated commercial system channels (ACSC).

4. CASE STUDY: AELCAC OPERATION SCENARIO

As an illustration, we present the AELCAC operation scenario using the example of the scheme shown in Fig. 2, which represents the city's power supply system. The generation includes a large cogeneration power plant with four units (60, 110, 110 and 124 MW) and four mini thermal power plants of 24 MW rated capacity each. Two units of large power plant with a nominal capacity of 110 MW each are under repair. The daily curve of the summer maximum load of the working day is considered, while the daily maximum load is 327.2 MW (Fig. 3).

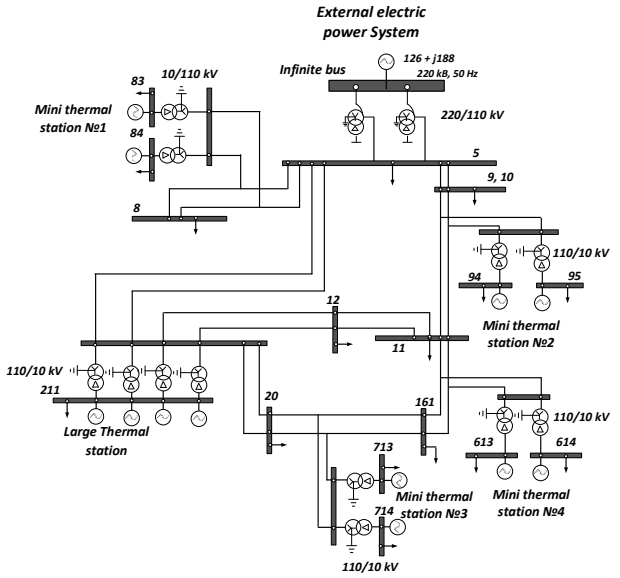


Figure 2. The studied power supply system.

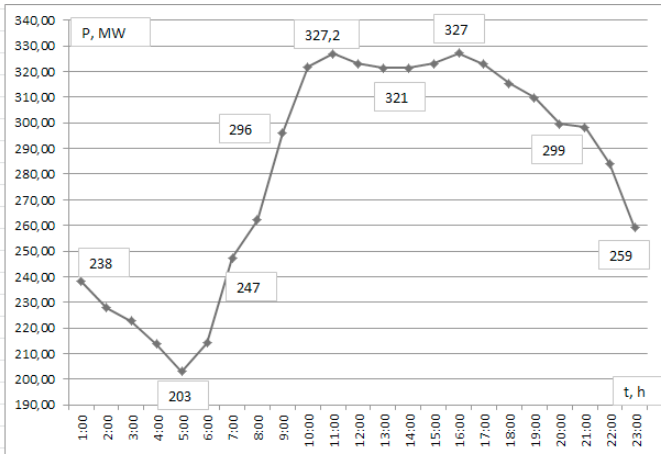


Figure 3. Daily load curve of power supply system.

Due to the false operation of protection devices at the 220-110 kV supply substation, an emergency shutdown of the 110 kV power lines connecting the PS with the external PS occurs. As a result, the PS is allocated for islanded operation with a power deficit of about 38% of the total power consumption.

Fig. 4 shows the dynamics of changes in the voltage and load power in one of the PS nodes, obtained because of calculating the transient process. At the time of 0.5 s, the PS was allocated for islanded operation with a power deficit of 126 MW and a sharp decrease in the voltage in the node. The power consumption in the node has decreased from 50 to 30 MW. The frequency reached the first UFLS set point, which triggered at 1.5 s, disabling 10% of the minor importance load. At the same time, the voltage level was partially restored, which, in accordance with the static voltage characteristics of the load, led to an increase in power consumption by the load to 43 MW. At time 1.8 s, the second UFLS set point triggered, disabling another 20% of the system load. As a result, by 2.0 s, the power consumed by the load in the node under consideration, taking into account the influence of static load characteristics and the action of automatic excitation regulators for power plant generators was set to about 36 MW.

At time 2.0 s, AELCAC worked, “moving” some electrical receivers with a total capacity of approximately 9MW in the node to the minimum of the load curve. In 2.3 s, the same automation connected previously disconnected consumers by about 6 MW.

At this point, the UFLS and AELCAC actions ended and the level of power consumed by the node load was set at approximately 36 MW. The frequency after some fluctuations was set at 49.8 Hz (Fig. 5). Fig. 6 shows a diagram of the UFLS operation for the case under study, which clearly demonstrates the operation of the AELCAC.

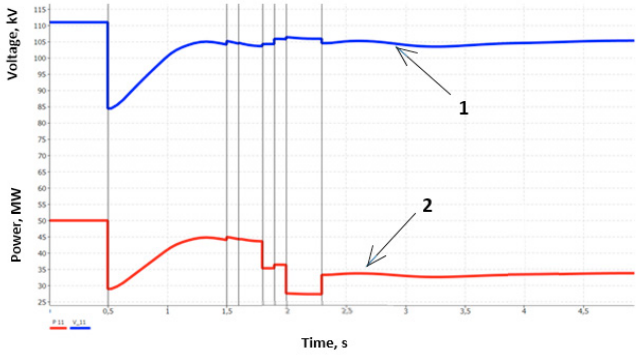


Figure 4. Dynamics of changes in voltage (1) and power consumption (2) in one of the PS nodes.

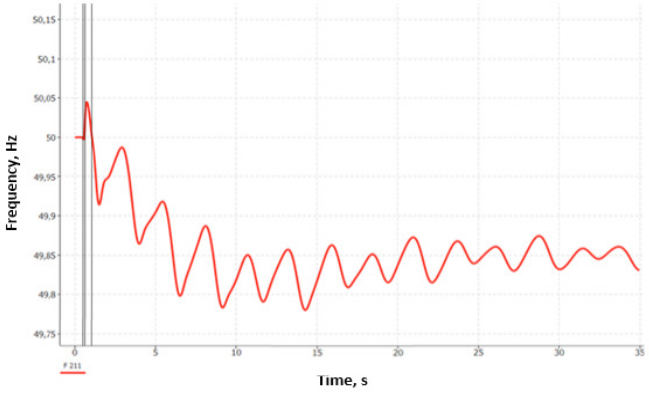


Figure 5. Dynamics of frequency changes in the system.

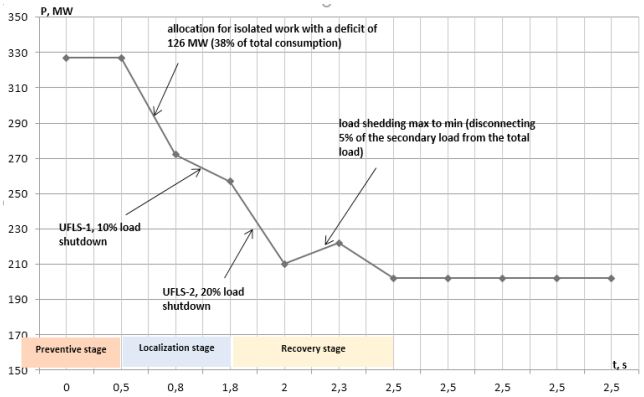


Figure 6. Diagram of the UFLS and AELCAC operation

5. SOME GENERALIZATIONS

The analysis of the principles of operation of various types of emergency load control automation and the directions of their modernization in accordance with current requirements allows us to draw the following generalizing conclusions.

1. An objectively determined trend for all types of emergency load shedding automation is the transfer of actuators for the implementation of CA to the level of the distribution network up to specific electrical receivers at a voltage of 0.4 kV. This makes it necessary to solve a new problem - monitoring overloads of elements of the distribution network due to the redistribution of power flows.

2. In fact, for all types of automatic emergency load shedding, the actuators for the implementation of CA use the same specific electric receivers. For example, an analysis of the situation in one of the areas of the electric distribution network of Irkutsk PS shows 95% coincidence of CA implementation sites under the operation of various types of emergency automation. This makes it necessary to coordinate the CA with the control of overloads of elements of the distribution network for various types of automatic emergency load shedding.

3. With this in mind, the problem of monitoring overloads of elements of the distribution network is common for all types of automatic equipment for emergency load shedding. It is necessary to develop an effective method for solving this problem. At the same time, it is necessary to detail the influencing factors, especially in terms of the category of electric receivers, features of the topology of the distribution network (radial or with closed loops), etc.

4. They require a detailed analysis of the possibility of using artificial intelligence methods, in particular artificial neural networks, to effectively solve the problems under consideration.

6. CONCLUSIONS

Real-time load control is an effective way to counteract the development of emergency processes in the PS and thereby increase the reliability of power supply to consumers. Therefore, many emergency control devices operate by controlling the load. Recently, there has been a tendency to move emergency load control actions from the level of 6-10 kV to 0.4 kV. This leads to the need to solve a new problem - monitoring overloads in the distribution electrical network. In addition, taking into account the implementation of control actions of different types of automation systems, as a rule, on the same electric receivers (feeders) it is necessary to coordinate the actions of various automatics, including the new proposed automation of emergency control of the load of active consumers.

AKNOWLEDGEMENTS

The work was carried out according to the state program № FWEU 2021-0001.

REFERENCES

Adamiak, M.G., Apostolov, A.P. and Begovic, M.M. (2006). Wide area protection – Technology and infrastructures. *IEEE Transactions on Power Delivery*, Vol.21, No. 3,601-609.

Aizenberg, N.I., Stashkevich, E.V. and Voropai, N.I. (2016). Coordination of cooperation of power supply company and active consumers in the optimization of daily load curves.*IzvestiyaRossijskojAkademiiNauk,Energetika*, No. 3, 1-11.(in Russian)

Conejo, A.J., Morales, J.M. and Baringo, L. (2017). Real time demand response model. *IEEE Transactions on Smart Grid*, Vol. 1, No. 3, 236-242.

Giyoev, B.M. and Artsyshevsky, Y.L. (2017). Under-frequency load shedding adaptation to variable level of electricity receivers responsibility on 0.4 kV. *Energetik*, No. 8, 18-21. (in Russian)

Faranda, R., Pievatolo, A. andTironi, E. (2007), Load shedding: A new proposal.*IEEE Transactions on Power Systems*, Vol. 22, No. 4, 2086-2093.

Danilov, M.V., Petrov, A.V. andChumakov, V.A. (2014). The structure of SALS adoptive system: Functions, organizational variants. *Scientific-Practical Conference "Power System: Properties Study, Control, Automatization"*, 147-151.Novosibirsk (in Russian)

Koshcheev, L.A. (2008). Development perspectives of automatic centralized system of emergency automation. *Elektricheskie Stantsii*, No. 2, 52-57. (in Russian)

Kulikov, A.L., llyushin, P.V.,Loskutov, A.A. and Sevostianov, A.A. (2021). Ensuring guaranteed high rapidity of under-frequency load shedding by deviations of electricity quality indexes using procedure of Wald successive analysis. *Elektricheskie Stantsii*, No. 4, 41-50. (in Russian)

Lu Jilingand Lin Lin. (2010). Dynamic load shedding based on transient energy margin sensitivity. *IEEE Power Con*, 6. China.

Panasetsky, D.A., Osak, A.B. and Voropai, N.I. (2011). An algorithm of automatic distributed load shedding for active power flow control, *International Scientific-Technical Conference "Modern Development Directions of Relay Protection and Automation Systems"*, 8. Saint-Petersburg. (in Russian)

Quint, R., Dangelmaier, L., Green, I. and Edelson, D. (2019). Transformation of the grid: The impact of distributed energy resources on bulk power systems, *IEEE Power and Energy Magazine*, Vol. 17, No. 6, 35-45.

Shulginov, N.G., Zhuko,v A.V., Dzemchuk, A.T. andKoshcheev, L.A. (2009). Emergency automatic control concept for Unified Energy System of Russia in current conditions, *Energetik*, No. 10, 2-4. (in Russian)

Sovalov, S.A. and Semenov, V.A. (1988).*Emergency control in power systems,* 416. Energoatomizdat, Moscow. (in Russian)

Stennikov, V.A., Barahtenko, E.A., Sokolov, D.V. and Shelekhova, V.B. (2017). Active demand-side management, *Izvestia vysshikh uchebnykh zanedeny. Problemy energetiki,* No. 11-12, 88-100. (in Russian)

Voropai, N.I., Efimov, D.N., Karataev, B.N., Novikov, E.A., Osak, A.B. andPanasetsky, D.A. (2016). Adoptive algorithms of automation for distributed load shedding,*ElektricheskieStantsii*, No. 11, 27-35. (in Russian)

Voropai, N.I. (2020) Electric power system transformations: A review of main prospects and challenges, *Energies*, 15.

Voropai, N.I. andChulyukova, M.V. (2020) Load emergency control for flexibility ensuring of electric power systems, *VestnikIrkutskogoGosudarstvennogoTehnicheskogoUniversiteta*, 22 (12), 57-168. (in Russian)

Ying-Yi, Ching-Yang Hsiao (2021). Event-based under-frequency load shedding scheme in a standalone power system, *Energies*, 14, 5659, 19